Return-Conservative Distributional Jump Bridges for Financial Disclosure Modeling

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Abstract

Financial disclosure models often let event text directly alter every predicted market-response target, including abnormal return. This is a flexible design, but it can destroy the weaker cross-sectional ranking signal carried by pre-event market state. We propose the Return-Conservative Distributional Jump Bridge (RC-DJB), a neural architecture that first estimates a no-event response distribution and then lets SEC filing text transport volatility, volume, and predictive uncertainty while imposing a hard firewall on the abnormal-return mean. The constraint is falsifiable: if disclosure text contains robust near-term return-ranking signal, blocking its direct return-mean path should hurt; if not, the bridge should improve event-response modeling without overwriting rank information. On a leakage-controlled public-data study with 7,236 SEC 8-K events, 7,236 matched controls, and 2,463 held-out rows from 98 large U.S. firms, RC-DJB obtains abnormal return rank IC 0.0546 with 95% ticker-clustered interval [0.0104, 0.0948]. It is the only tested model that both beats concat fusion on multi-target MSE by a positive paired interval (0.0043, [0.0023, 0.0061]) and has a strictly positive ticker-clustered rank-IC interval. Event bridge interventions increase held-out event-row MSE by 0.0066, 0.0049, and 0.0081 when the bridge is removed, text is zeroed, or text is shuffled. The result does not establish a complete trading system; it supports a sharper modeling claim: disclosure text is useful for response-distribution transport, but directly using it to move the return mean is harmful in this setting.

1 Introduction

Public disclosures are timed interventions in the market information set. A filing can change volatility, liquidity, and uncertainty even when its effect on cross-sectional abnormal-return ranking is weak or unstable. Standard multimodal forecasters typically concatenate price history and text embeddings, then predict all targets with a shared head. This design asks the same text representation to explain risk response, volume response, uncertainty, and return ranking. In financial data, that coupling is a liability: return labels are noisy, text is high dimensional, and small leakage or overfitting effects can look like alpha.

This paper studies a stricter architectural hypothesis. SEC disclosure text should be allowed to transport the event-response distribution, but it should not directly overwrite the mean abnormal return predicted from the no-event market state. We call the resulting model a Return-Conservative Distributional Jump Bridge (RC-DJB). The model first predicts a no-event Gaussian response distribution from pre-event price, volume, and market-relative features. It then uses disclosure text to generate bounded shifts in volatility-jump mean, volume-jump mean, and predictive log-variance. The abnormal-return mean shift is constrained to zero.

The contributions are:

1. A return-conservative distributional bridge architecture for event-driven financial modeling.
2. A leakage-controlled SEC 8-K experiment with matched no-event controls and public post-filing market outcomes.
3. Evidence that blocking direct text-to-return-mean transport recovers statistically positive rank IC while preserving response-regression gains over fusion baselines.

4. Intervention tests showing that the disclosure bridge contributes to event-row MSE even though return ranking is intentionally anchored to the no-event state.

2 Related Work

Classical event studies estimate abnormal returns around information releases [Fama et al., 1969, Brown and Warner, 1985, MacKinlay, 1997]. RC-DJB keeps this information boundary but replaces scalar event-window estimation with a neural distributional response model. The architecture is related to state-space and dynamical-system methods, including Koopman operators [Koopman, 1931], neural differential equations [Chen et al., 2018], structured state-space sequence models [Gu et al., 2022], and selective state-space models [Gu and Dao, 2023]. Marked point processes model event arrivals [Hawkes, 1971, Mei and Eisner, 2017]; here the disclosure time is observed and the task is post-event response prediction.

Financial text contains information relevant to prices and fundamentals [Loughran and McDonald, 2011, Cohen et al., 2020, Araci, 2019]. Recent work studies multimodal financial forecasting, financial reasoning benchmarks, and leakage in financial agents [Chen et al., 2025, Islam et al., 2023, Jiang et al., 2026, Li et al., 2025]. Causal Transformer and G-Transformer estimate counterfactual outcomes in temporal settings [Melnychuk et al., 2022, Feng et al., 2024]. RC-DJB differs by imposing a target-specific architectural restriction: disclosure text may transport response distribution components, but the abnormal-return mean remains no-event anchored.

3 Method

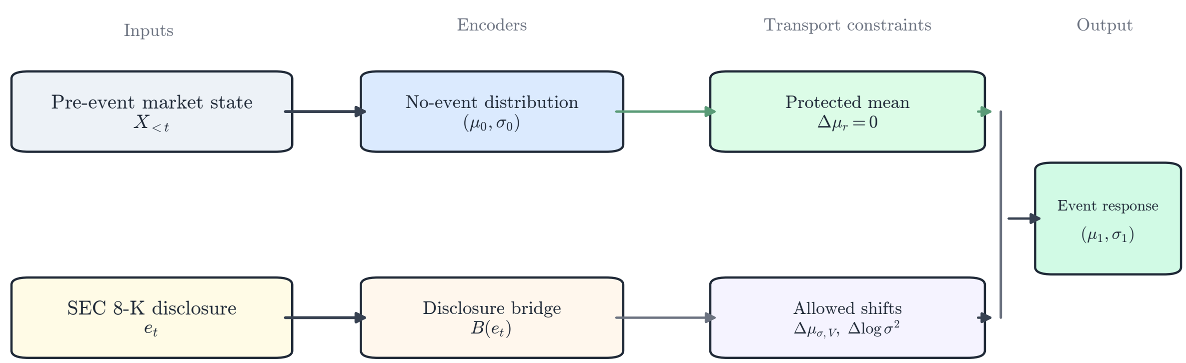

Figure 1: RC-DJB first estimates a no-event response distribution from market history. Disclosure text generates a bounded bridge for risk/liquidity response and predictive uncertainty, but a return-mean firewall enforces $\Delta\mu_r = 0$.

Inputs and targets. Each sample i has label start date τ_i, a 40-trading-day pre-event tensor $X_i \in \mathbb{R}^{40\times 8}$, disclosure embedding e_i, metadata $m_i \in \mathbb{R}^6$, and target vector $y_i \in \mathbb{R}^3$. The targets are five-trading-day SPY-adjusted return, realized-volatility jump, and volume jump. The headline variant uses chunk-pooled BGE-small disclosure embeddings [Xiao et al., 2023].

No-event distribution. A recurrent encoder maps pre-event market history to a latent state:

$$h_i = \mathrm{GRU}_\theta(X_i) + W_m m_i. \tag{1}$$

A bounded gate produces the no-event state $\tilde{s}_i$, from which the model predicts a diagonal Gaussian distribution:

$$\tilde{s}_i = h_i \odot (1 + 0.1 \tanh(g_\theta(h_i, m_i))), \tag{2}$$
$$[\mu_{0,i}, \ell_{0,i}] = f_\theta(\tilde{s}_i, m_i), \tag{3}$$

where $\ell_{0,i}$ is log-variance.

Distributional bridge. Disclosure text generates a bounded latent state jump and distributional transport:

$$b_i = B_\theta(e_i), \tag{4}$$
$$j_i = \alpha \tanh(u_\theta(b_i)) \odot \sigma(v_\theta(b_i)), \tag{5}$$
$$[\Delta\mu_i, \Delta\ell_i] = T_\theta(\mathrm{LN}(\tilde{s}_i + j_i), b_i, m_i). \tag{6}$$

The central architectural constraint is

$$\Delta\mu_{i,return} = 0. \tag{7}$$

Thus,

$$\mu_{1,i} = \mu_{0,i} + [0, \Delta\mu_{i,volatility}, \Delta\mu_{i,volume}], \qquad \ell_{1,i} = \mathrm{clip}(\ell_{0,i} + \Delta\ell_i). \tag{8}$$

The model predicts $p_\theta(y_i) = \mathcal{N}(\mu_{1,i}, \mathrm{diag}(\exp(\ell_{1,i})))$.

Objective. The main loss combines target-standardized Huber regression, Gaussian negative log likelihood, control suppression, bridge sparsity, residual consistency, and a pairwise rank regularizer:

$$\mathcal{L} = \lambda_y \mathcal{L}_{Huber}(D_y^{-1}(\mu_1 - y); w_y) + \lambda_n \mathcal{L}_{NLL} + \lambda_c \mathbb{1}_{control} \|\Delta\mu_i\|_2^2 + \lambda_s \|j_i\|_1 + \lambda_v \mathrm{Var}(\Delta\mu_i) + \lambda_r \mathcal{L}_{rank}. \tag{9}$$

The main run uses $w_y = (3.0, 1.0, 0.25)$, $\lambda_n = 0.10$, $\lambda_c = 0.35$, $\lambda_s = 0.001$, $\lambda_v = 0.05$, and $\lambda_r = 0.15$.

4 Data and Leakage Controls

The experiment uses public SEC EDGAR APIs [U.S. Securities and Exchange Commission, 2026]. The sample covers 98 usable large U.S. firms, SEC 8-K filings from 2020–2025, and public daily prices. Tickers are mapped to CIKs through the SEC company-tickers file. Filings are read from SEC submissions JSON, including archived submissions, using `acceptanceDateTime` and the primary EDGAR document URL. The parser stores source URL, accession number, accepted timestamp, available timestamp, extracted text, and text hash.

Daily prices are downloaded from public price endpoints, with SPY used as the market proxy for abnormal returns. The trading calendar is inferred from observed price rows. Event label windows start from the first tradable session after public availability; filings accepted after 16:00 ET or on non-trading days start on the next observed trading day. Features use only prices strictly before the label start date. Labels use future returns only as targets. One deterministic same-ticker no-event control is constructed for each filing, excluding dates within a 10-calendar-day blackout window around known events.

Table 1: Main public-data sample.

Artifact	Count
Usable companies	98
SEC 8-K events	7,236
Matched no-event controls	7,236
Feature rows	14,472
Public price rows	159,093
Held-out rows	2,463

5 Experiments

Rows with $\tau_i \leq$ 2023-12-31 are training rows, rows from 2024 are validation rows, and later rows are held out. This yields 9,729 training rows, 2,280 validation rows, and 2,463 held-out rows across 98 held-out tickers and 13 held-out label-start months. The headline RC-DJB checkpoint is selected by validation loss. All models use fixed seed 29 and are evaluated once on the held-out split.

Baselines include no-event dynamics, text-only MLP, concat fusion, Counterfactual Event Bottleneck Transformer (CEBT), Disclosure Operator Transformer (DOT), Event-Jump State Space Model (EJSSM), and an unconstrained Distributional Jump Bridge (DJB). Confidence intervals use percentile bootstrap resampling; ticker-clustered intervals resample tickers with replacement. Paired comparisons use row-paired bootstrap resampling. Rank IC is Spearman correlation between predicted and realized abnormal return.

Table 2: Held-out results. MSE is over abnormal return, volatility jump, and volume jump. Intervals are 95% ticker-clustered bootstrap intervals.

Model	MSE	MSE CI	Rank IC	Rank IC CI
No-event	0.0696	[0.0519, 0.0936]	-0.0411	[-0.0840, 0.0016]
Concat fusion	0.0597	[0.0442, 0.0817]	-0.0182	[-0.0597, 0.0267]
CEBT	0.0660	[0.0488, 0.0898]	0.0463	[0.0052, 0.0893]
DOT	0.0596	[0.0449, 0.0804]	-0.0131	[-0.0510, 0.0291]
EJSSM-BGE	0.0547	[0.0395, 0.0764]	0.0099	[-0.0329, 0.0556]
DJB	**0.0539**	[0.0389, 0.0757]	0.0128	[-0.0290, 0.0547]
RC-DJB	0.0554	[0.0400, 0.0776]	**0.0546**	[0.0104, 0.0948]

Table 3: Paired comparisons against RC-DJB. Positive MSE values mean RC-DJB has lower MSE. Positive rank values mean RC-DJB has higher rank IC.

Baseline	MSE Diff.	95% CI	Rank IC Diff.	95% CI
No-event	0.01417	[0.01150, 0.01747]	0.0956	[0.0412, 0.1504]
Concat fusion	0.00427	[0.00230, 0.00612]	0.0728	[0.0197, 0.1249]
CEBT	0.01057	[0.00825, 0.01311]	0.0083	[-0.0500, 0.0668]
DOT	0.00418	[0.00180, 0.00638]	0.0677	[0.0186, 0.1180]
EJSSM-BGE	-0.00076	[-0.00163, 0.00005]	0.0447	[-0.0072, 0.0952]

Table 4: RC-DJB diagnostics on the held-out split.

Diagnostic	Value
Gaussian NLL	-2.2864
Observed 80% interval coverage	0.9011
Observed 95% interval coverage	0.9602
Mean predictive standard deviation	0.1302
Mean response transport, real 8-Ks	0.0358
Mean response transport, controls	0.0073
Latent bridge AUC, events vs. controls	0.6038

Table 5: RC-DJB intervention tests. Rank IC is unchanged by bridge interventions because the return-mean firewall fixes the abnormal-return point prediction to the no-event mean.

Variant	All MSE	Event MSE	Event NLL	Rank IC	Bridge AUC
Full RC-DJB	**0.0554**	**0.0843**	**-2.2115**	0.0546	0.6038
No bridge	0.0586	0.0909	-2.1204	0.0546	0.5000
Zero text	0.0578	0.0892	-2.1886	0.0546	0.4843
Shuffled text	0.0597	0.0924	-2.1765	0.0546	0.5057

Table 6: Paired event-row MSE stress tests. Positive values mean full RC-DJB has lower event-row MSE than the intervention.

Intervention	Event MSE increase	95% paired CI
No bridge	0.0066	[0.0043, 0.0093]
Zero text	0.0049	[0.0027, 0.0073]
Shuffled text	0.0081	[0.0051, 0.0117]

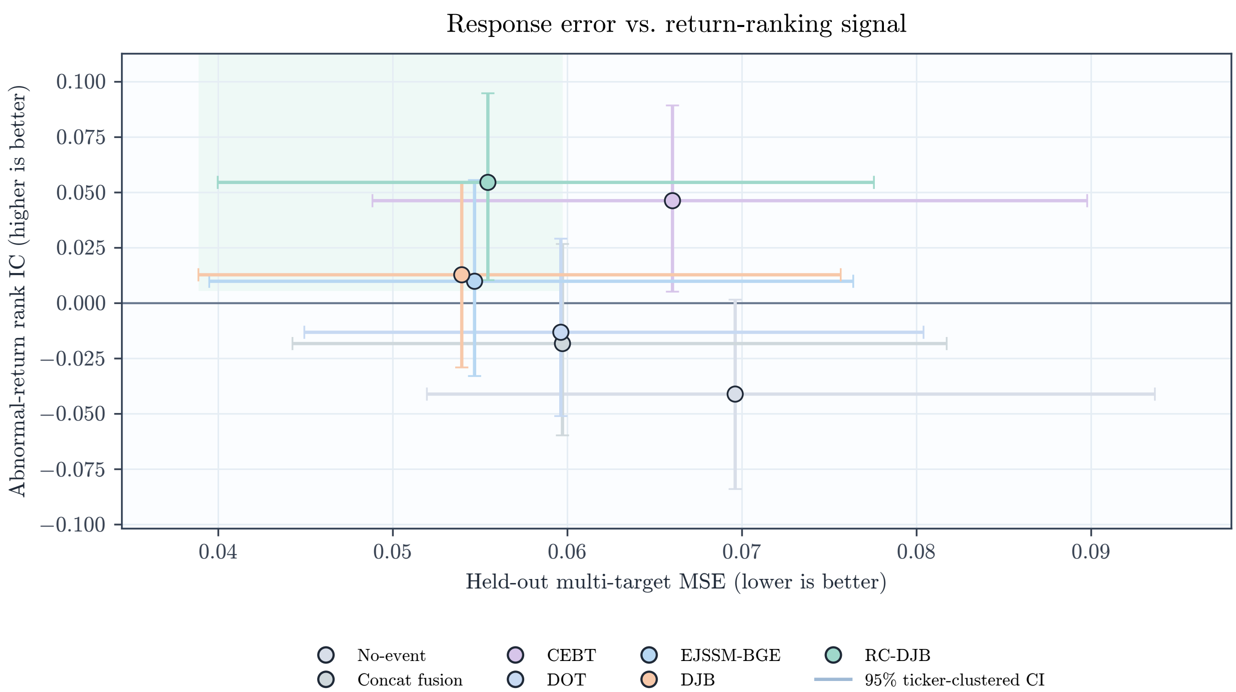

Figure 2: RC-DJB occupies a different point on the response/ranking tradeoff: EJSSM and DJB have lower point MSE, while RC-DJB is the only tested low-error model with a strictly positive ticker-clustered rank-IC interval.

Figure 3: Interventions show that the disclosure bridge matters for event-response error. Rank IC does not move under text interventions because the return-mean firewall prevents direct text-to-return overwriting.

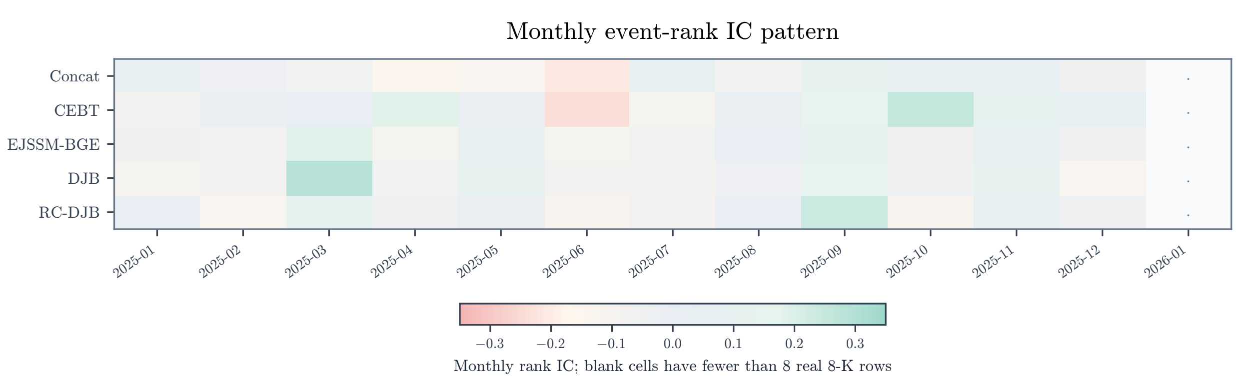

Figure 4: Monthly event-rank IC over held-out real 8-K rows. The model-level rank result is not uniform across months, but RC-DJB has a more positive held-out pattern than fusion and unconstrained event-mean transport baselines.

6 Findings

Return-mean conservation recovers rank signal. RC-DJB obtains rank IC 0.0546 with ticker-clustered interval [0.0104, 0.0948]. In contrast, concat fusion, DOT, EJSSM-BGE, and unconstrained DJB all have rank intervals crossing zero. CEBT also has positive rank IC, but its MSE is much worse than RC-DJB.

The bridge improves response modeling without moving return means. RC-DJB improves paired MSE over concat fusion by 0.0043 and over DOT by 0.0042. Removing the bridge raises event-row MSE by 0.0066; zeroing or shuffling text raises event-row MSE by 0.0049 and 0.0081. These changes occur even though abnormal-return predictions are unchanged by construction, which means the bridge is acting through volatility, volume, and uncertainty transport.

There is a real response/ranking tradeoff. EJSSM-BGE and unconstrained DJB achieve lower point MSE than RC-DJB, but their rank IC intervals include zero. RC-DJB is therefore not a universal winner; it is a Pareto-frontier point that prioritizes statistically reliable ranking while keeping response error below fusion baselines. This tradeoff is the main empirical evidence for the return firewall.

The likelihood result is competitive but not decisive. RC-DJB has Gaussian NLL -2.2864, close to EJSSM-BGE's -2.2839. The paired NLL difference against EJSSM-BGE is small and not statistically decisive. The likelihood evidence supports the bridge as competitive; the distinctive result is the combination of positive rank IC and response-error improvement over fusion baselines.

7 Reproducibility

The artifact records configurations, seeds, processed event metadata, text hashes, price rows, feature metadata, checkpoints, predictions, table CSVs, and figure outputs used for the reported results. Tests cover timestamp gating, after-hours label starts, feature leakage rejection, control construction, deterministic cached tensors, finite model losses, probabilistic loss integration, and evaluation leakage guards. Labels are computed only from future price windows and are never included in model inputs.

8 Limitations

The study has several limitations. The universe is large-cap biased and may contain survivorship effects, ticker-history issues, and 8-K item-mix sensitivity. The price source is public daily data rather than CRSP or intraday trades and quotes. Abnormal returns are SPY-adjusted rather than factor-model residuals. BGE-small is a general-purpose encoder, and filings are represented by chunk mean pooling rather than a trained long-document financial encoder. Rank IC is statistically positive but economically small; transaction costs, capacity, and portfolio construction are outside the scope of this study. The return firewall is an architectural and predictive constraint, not structural causal identification.

9 Ethics and Use

This paper studies event representations using public filings and public daily prices. The results do not constitute trading recommendations. The artifact is intended to support leakage auditing, independent reproduction, and falsification of the empirical claims.

10 Conclusion

RC-DJB shows that the route by which disclosure text enters a financial event model matters. Letting text directly move the abnormal-return mean is not necessary for response modeling and appears to harm ranking in this setting. A return-conservative distributional bridge obtains positive held-out rank IC while still improving multi-target event-response MSE over concat fusion and DOT. The evidence supports a more precise architecture principle for disclosure modeling: use text to transport response risk, liquidity, and uncertainty, but keep near-term return ranking anchored to the no-event market state unless stronger evidence justifies crossing that boundary.

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